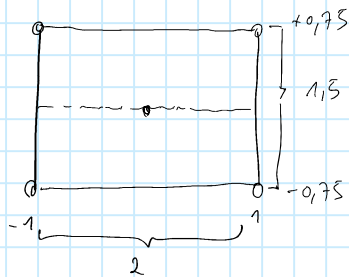


Ü-Projektion-WS24/25-Muster

1a)

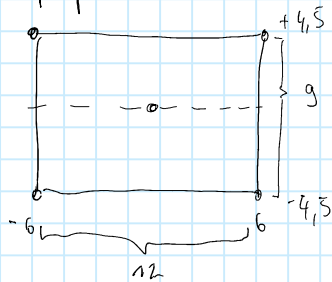
near plane:



$$4:3 \Rightarrow \frac{2}{x} = \frac{4}{3}$$

$$1,5 = x$$

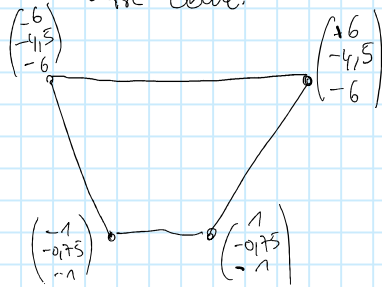
far plane



$$4:3 \Rightarrow \frac{12}{x} = \frac{4}{3}$$

$$9 = x$$

unter Ebene:



links-rechts : $l = -1$

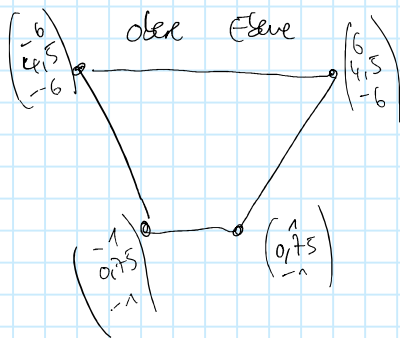
(x-Ebene) $r = +1$

oben-unten : $b = -0,75$

(y-Ebene) $t = 0,75$

Vorne-hinten : $f = -6$

(z-Ebene) $n = -1$



Matrix aufstellen:

$$\begin{pmatrix} \frac{2 \cdot (-1)}{1+1} & 0 & \frac{-1+1}{1+1} & 0 \\ 0 & \frac{2 \cdot (-1)}{0,75+0,75} & \frac{-0,75+0,75}{0,75+0,75} & 0 \\ 0 & 0 & -\frac{-6-1}{-6+1} & \frac{2 \cdot (-6) \cdot (-1)}{-6+1} \\ 0 & 0 & +1 & 0 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -4/3 & 0 & 0 \\ 0 & 0 & -7/5 & -12/5 \\ 0 & 0 & +1 & 0 \end{pmatrix}$$

$$P_{\text{new}} = M \cdot P_{\text{alt}}$$

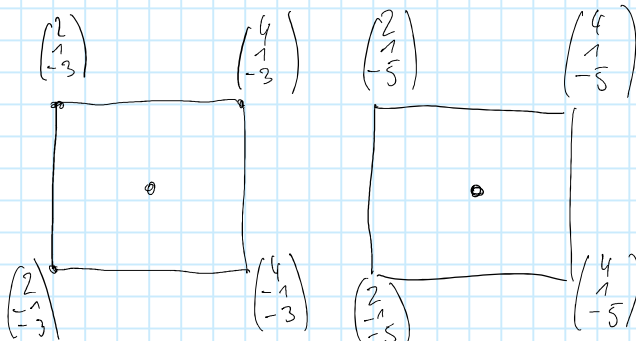
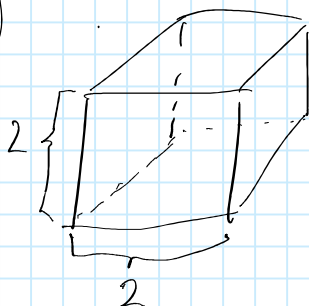
far, right, top

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -4/3 & 0 & 0 \\ 0 & 0 & -7/5 & -12/5 \\ 0 & 0 & +1 & 0 \end{pmatrix} \begin{pmatrix} 6 \\ 4.5 \\ -6 \\ 1 \end{pmatrix} = \begin{pmatrix} -6 \\ -6 \\ 6 \\ 6 \end{pmatrix}$$

Division durch w :

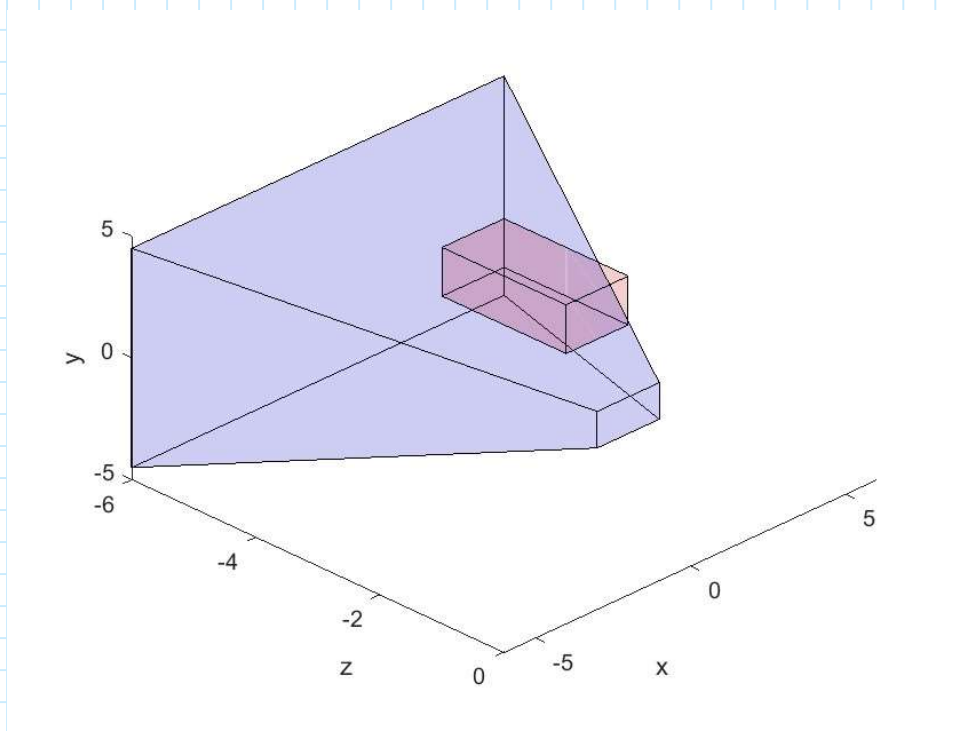
$$\begin{pmatrix} -6 \\ -6 \\ 6 \\ -6 \end{pmatrix} \cdot w = \begin{pmatrix} +1 \\ +1 \\ -1 \\ 1 \end{pmatrix}$$

2a)



Matrix und Berechnung der transf. Punkte wie in A1

Start-Situation:



End-Situation

